

VPC Networking Fundamentals – Cloud Network Security & Traffic Analysis

Project Overview

Cloud networking forms the backbone of cloud security. In this project, I designed and analyzed a **Virtual Private Cloud (VPC)** using **Google Cloud Platform (GCP)** to understand how cloud networks are structured, how firewall rules control traffic, and how internal and external communication behaves in a secure cloud environment.

This project emphasizes **hands-on cloud networking and security fundamentals**, including firewall rule analysis, internal vs external traffic testing, and basic network troubleshooting, using **free Google Cloud labs**.

Project Objectives

The objectives of this project were to:

- Design and configure a cloud-based virtual network (VPC)
- Deploy and manage virtual machines within the network
- Analyze and understand cloud firewall rules
- Test and compare internal (private) and external (public) network traffic
- Build practical foundational skills in cloud networking and security

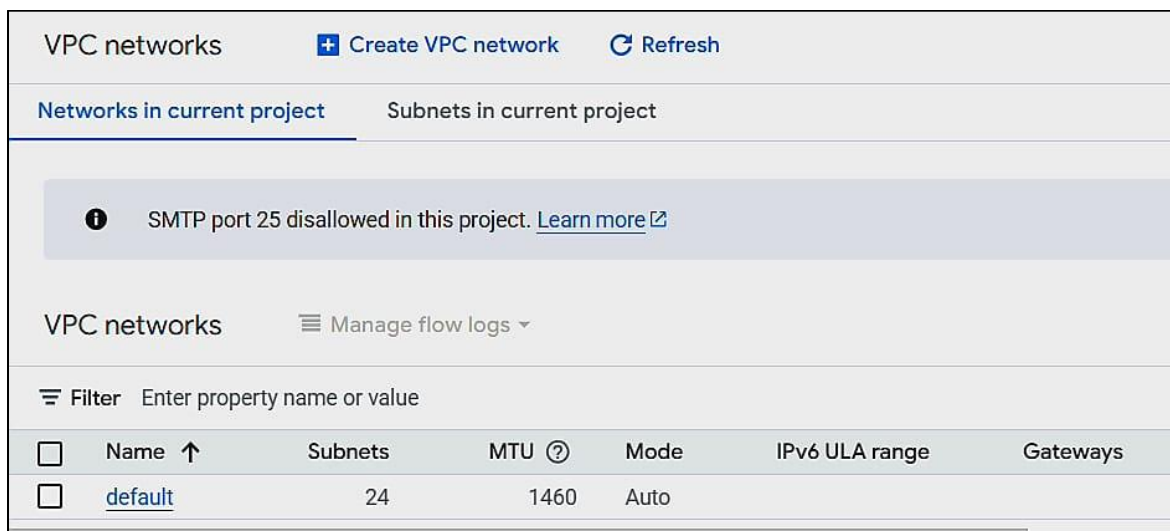
Tools & Technologies Used

- **Google Cloud Platform (GCP)**
- **VPC Networks**
- **Compute Engine** (Virtual Machines)
- **Firewall Rules** (Ingress traffic control)
- **Linux Command Line** (ping, SSH)

Project Architecture Overview

The project architecture consists of:

- One **auto-mode VPC network** (automatically creates subnets)
- Two **Compute Engine virtual machine instances**
- Default firewall rules controlling:
 - ICMP (ping)
 - SSH (TCP port 22)
- Network testing using both **internal and external IP addresses**



VPC networks							+ Create VPC network	Refresh
Networks in current project							Subnets in current project	
SMTP port 25 disallowed in this project. Learn more								
VPC networks							Manage flow logs	
Filter Enter property name or value								
☐	Name ↑	Subnets	MTU ?	Mode	IPv6 ULA range	Gateways		
☐	default	24	1460	Auto				

Fig. 1. VPC Network overview page

Step 1: VPC Network Creation

An **auto-mode VPC network** was created to quickly provision subnets across multiple regions. This helped me understand how Google Cloud simplifies network creation while still enforcing secure defaults.

Key Observations:

- Subnets are automatically created for each region
- Each subnet is assigned a private IP range
- Default firewall rules are applied at the network level

Networks in current projectSubnets in current project

Subnets

manage view logs

Filter

Enter property name or value

☐	Name	Region	VPC network ↑	IP stack type	Primary IPv4 range	Secondary IPv4 ranges	IPv6 ranges	Gateways
☐	default	africa-south1	default	IPv4	10.218.0.0/20			10.218.0.1
☐	default	asia-east1	default	IPv4	10.140.0.0/20			10.140.0.1
☐	default	asia-south1	default	IPv4	10.160.0.0/20			10.160.0.1
☐	default	asia-southeast1	default	IPv4	10.148.0.0/20			10.148.0.1
☐	default	asia-southeast3	default	IPv4	10.232.0.0/20			10.232.0.1
☐	default	australia-southeast1	default	IPv4	10.152.0.0/20			10.152.0.1
☐	default	europa-central2	default	IPv4	10.186.0.0/20			10.186.0.1
☐	default	europa-north2	default	IPv4	10.226.0.0/20			10.226.0.1
☐	default	europa-west1	default	IPv4	10.132.0.0/20			10.132.0.1
☐	default	europa-west10	default	IPv4	10.214.0.0/20			10.214.0.1
☐	default	europa-west12	default	IPv4	10.210.0.0/20			10.210.0.1

Fig. 2. VPC subnet details

Step 2: Virtual Machine Deployment

Two virtual machines were deployed within the same VPC network to test network connectivity and security controls.

Each VM was assigned:

- An **internal IP address** for private, intra-network communication
- An **external IP address** for public access and testing

VM instances						
Filter Enter property name or value						
Name ↑	Zone	Recommendations	In use by	Internal IP	External IP	Connect
vm-1	us-east1-c			10.142.0.2 (nic0)	35.185.14.163 (nic0)	SSH ▾ ⋮
vm-2	us-east1-c			10.142.0.3 (nic0)	34.23.165.49 (nic0)	SSH ▾ ⋮

Fig. 3. VM instances page

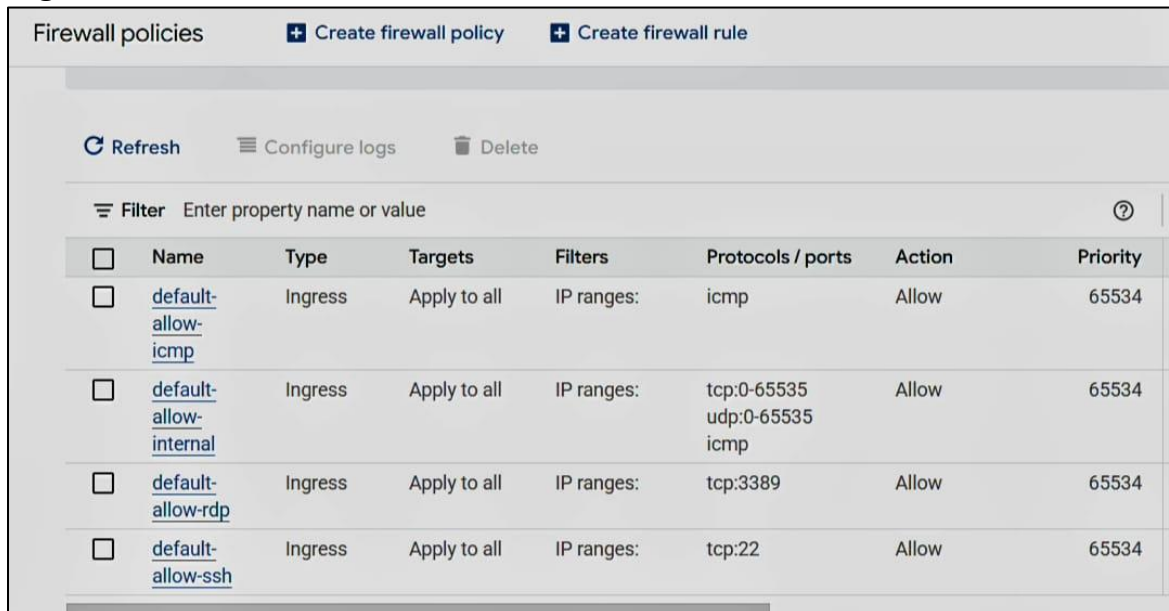
Step 3: Firewall Rules Analysis

Google Cloud uses **stateful firewall rules** to control ingress and egress traffic. I analyzed the default firewall rules associated with the VPC, including:

- ICMP rule (allows ping traffic)
- SSH rule (allows TCP port 22)
- Implicit deny rules for unspecified traffic

Key Learnings:

- Firewall rules are evaluated based on **priority**
- Rules are applied at the **VPC (network) level**
- Ingress rules determine what traffic is allowed into instances



Firewall policies							
+ Create firewall policy + Create firewall rule							
Refresh Configure logs Delete							
Filter Enter property name or value ?							
☐	Name	Type	Targets	Filters	Protocols / ports	Action	Priority
☐	default-allow-icmp	Ingress	Apply to all	IP ranges:	icmp	Allow	65534
☐	default-allow-internal	Ingress	Apply to all	IP ranges:	tcp:0-65535 udp:0-65535 icmp	Allow	65534
☐	default-allow-rdp	Ingress	Apply to all	IP ranges:	tcp:3389	Allow	65534
☐	default-allow-ssh	Ingress	Apply to all	IP ranges:	tcp:22	Allow	65534

Fig. 4. Firewall rules list

Step 4: Internal Traffic Testing (East-West Traffic)

To test internal communication, I connected to one virtual machine and used the ping command to reach the **internal IP address** of the second virtual machine.

ping <internal-ip>

Result:

- Internal traffic was successful
- Communication remained within the private VPC network
- No public internet routing was involved

This demonstrated **east–west traffic behavior**, which is commonly used for communication between services inside cloud environments.

```
SSH-in-browser

Linux vm-1 6.1.0-42-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.159-1 (2025-12-30) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Creating directory '/home/student-00-d0e591c0bece'.
student-00-d0e591c0bece@vm-1:~$ ping vm-2
PING vm-2.us-east1-c.c.qwiklabs-gcp-00-3efabebbe55f.internal (10.142.0.3) 56(84) bytes of data.
64 bytes from vm-2.us-east1-c.c.qwiklabs-gcp-00-3efabebbe55f.internal (10.142.0.3): icmp_seq=1 ttl=64 time=1.49 ms
64 bytes from vm-2.us-east1-c.c.qwiklabs-gcp-00-3efabebbe55f.internal (10.142.0.3): icmp_seq=2 ttl=64 time=0.254 ms
64 bytes from vm-2.us-east1-c.c.qwiklabs-gcp-00-3efabebbe55f.internal (10.142.0.3): icmp_seq=3 ttl=64 time=0.292 ms
64 bytes from vm-2.us-east1-c.c.qwiklabs-gcp-00-3efabebbe55f.internal (10.142.0.3): icmp_seq=4 ttl=64 time=0.238 ms
64 bytes from vm-2.us-east1-c.c.qwiklabs-gcp-00-3efabebbe55f.internal (10.142.0.3): icmp_seq=5 ttl=64 time=0.309 ms
64 bytes from vm-2.us-east1-c.c.qwiklabs-gcp-00-3efabebbe55f.internal (10.142.0.3): icmp_seq=6 ttl=64 time=0.261 ms
64 bytes from vm-2.us-east1-c.c.qwiklabs-gcp-00-3efabebbe55f.internal (10.142.0.3): icmp_seq=7 ttl=64 time=0.211 ms
64 bytes from vm-2.us-east1-c.c.qwiklabs-gcp-00-3efabebbe55f.internal (10.142.0.3): icmp_seq=8 ttl=64 time=0.265 ms
64 bytes from vm-2.us-east1-c.c.qwiklabs-gcp-00-3efabebbe55f.internal (10.142.0.3): icmp_seq=9 ttl=64 time=0.246 ms
64 bytes from vm-2.us-east1-c.c.qwiklabs-gcp-00-3efabebbe55f.internal (10.142.0.3): icmp_seq=10 ttl=64 time=0.228 ms
64 bytes from vm-2.us-east1-c.c.qwiklabs-gcp-00-3efabebbe55f.internal (10.142.0.3): icmp_seq=11 ttl=64 time=0.240 ms
64 bytes from vm-2.us-east1-c.c.qwiklabs-gcp-00-3efabebbe55f.internal (10.142.0.3): icmp_seq=12 ttl=64 time=0.328 ms
64 bytes from vm-2.us-east1-c.c.qwiklabs-gcp-00-3efabebbe55f.internal (10.142.0.3): icmp_seq=13 ttl=64 time=0.245 ms
64 bytes from vm-2.us-east1-c.c.qwiklabs-gcp-00-3efabebbe55f.internal (10.142.0.3): icmp_seq=14 ttl=64 time=0.232 ms
64 bytes from vm-2.us-east1-c.c.qwiklabs-gcp-00-3efabebbe55f.internal (10.142.0.3): icmp_seq=15 ttl=64 time=0.232 ms
64 bytes from vm-2.us-east1-c.c.qwiklabs-gcp-00-3efabebbe55f.internal (10.142.0.3): icmp_seq=16 ttl=64 time=0.284 ms
```

```
SSH-in-browser

Linux vm-2 6.1.0-42-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.159-1 (2025-12-30) x86_64

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the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Fri Jan 16 16:37:52 2026 from 35.235.243.209
student-00-d0e591c0bece@vm-2:~$ ping 10.142.0.2
PING 10.142.0.2 (10.142.0.2) 56(84) bytes of data.
64 bytes from 10.142.0.2: icmp_seq=1 ttl=64 time=1.50 ms
64 bytes from 10.142.0.2: icmp_seq=2 ttl=64 time=0.217 ms
64 bytes from 10.142.0.2: icmp_seq=3 ttl=64 time=0.228 ms
64 bytes from 10.142.0.2: icmp_seq=4 ttl=64 time=0.293 ms
64 bytes from 10.142.0.2: icmp_seq=5 ttl=64 time=0.241 ms
64 bytes from 10.142.0.2: icmp_seq=6 ttl=64 time=0.310 ms
```

Fig. 5. and 6. Successful internal IP ping

Step 5: External Traffic Testing (North–South Traffic)

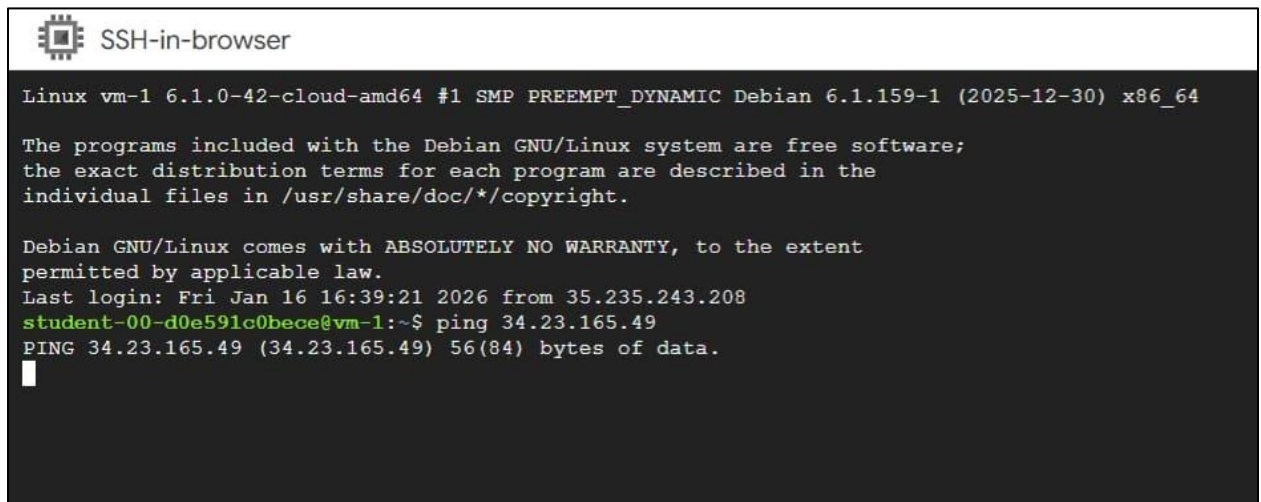
Next, I tested connectivity using the **external IP address** of the second virtual machine.

This test demonstrated:

- **North–south traffic behavior** (traffic entering or leaving the cloud network)
- Dependency on firewall rules for public access
- Higher exposure compared to internal communication

Key Learning:

- Internal IP traffic is more secure and efficient
- External IP traffic requires stricter firewall and security controls

A screenshot of an SSH-in-browser terminal window. The title bar shows a gear icon and the text "SSH-in-browser". The terminal output shows the login banner for a Debian 6.1.159-1 VM, followed by a successful ping command to the external IP 34.23.165.49. The ping result shows 56(84) bytes of data.

```
Linux vm-1 6.1.0-42-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.159-1 (2025-12-30) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Fri Jan 16 16:39:21 2026 from 35.235.243.208
student-00-d0e591c0bece@vm-1:~$ ping 34.23.165.49
PING 34.23.165.49 (34.23.165.49) 56(84) bytes of data.

```

Fig. 7. External IP connectivity test

Step 6: Security Observations & Analysis

Based on the testing and analysis performed:

- Internal traffic is faster and inherently more secure

- Firewall rules are critical for controlling access to cloud resources
- Google Cloud firewalls are **stateful by default**
- Public exposure of resources should always be minimized following **least-privilege networking principles**

Security Concepts Demonstrated

- Cloud network isolation using VPCs
- Firewall rule enforcement and traffic control
- Internal vs external traffic analysis
- Secure cloud network design fundamentals

Skills Gained

- Cloud networking fundamentals
- Google Cloud VPC configuration
- Firewall rule analysis and troubleshooting
- Understanding of east–west and north–south traffic
- Basic Linux networking commands
- Security-first thinking in cloud environments

Project Deliverables

- Configured Google Cloud VPC network
- Two operational virtual machine instances
- Verified internal and external traffic behavior
- Screenshots documenting architecture and testing